International Financial Statement Analysis

Chapter 3: Understanding Income Statements – Part 1

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Learning Outcomes

- Describe the components of the income statement and alternative presentation formats of that statement;
- Describe general principles of revenue recognition and accounting standards for revenue recognition;
- Calculate revenue given information that might influence the choice of revenue recognition method;
- Describe general principles of expense recognition, specific expense recognition applications, and implications of expense recognition choices for financial analysis;
- Describe the financial reporting treatment and analysis of non-recurring items (including discontinued operations, unusual or infrequent items) and changes in accounting policies;
- Distinguish between the operating and non-operating components of the income statement;

Learning Outcomes (cont’d)

- Describe how earnings per share is calculated and calculate and interpret a company's earnings per share (both basic and diluted earnings per share) for both simple and complex capital structures;
- Distinguish between dilutive and antidilutive securities and describe the implications of each for the earnings per share calculation;
- Convert income statements to common-size income statements;
- Evaluate a company's financial performance using common-size income statements and financial ratios based on the income statement;
- Describe, calculate, and interpret comprehensive income;
- Describe other comprehensive income and identify major types of items included in it.

Introduction

- The income statement presents information on the financial results of a company's business activities over a period of time.
- The basic equation underlying the income statement, ignoring gains and losses, is

Revenue - Expenses = Net income

- Sometimes referred to as the “statement of operations,” “statement of earnings,” or “profit and loss (P&L) statement.”
- Under both IFRS and U.S. GAAP, the income statement may be presented as a separate statement followed by a statement of comprehensive income that begins with the profit or loss from the income statement or as a section of a single statement of comprehensive income.

Introduction (cont’d)

- Equity analysts are interested in income statements because inputs into valuation models often include estimates of earnings.
- Fixed-income analysts examine the components of income statements, past and projected, to assess companies' ability to make promised payments on their debt.

Components and Format of the Income Statement

	12 Months Ended December 31		
	2017	2016	2015
Revenue	$56,444	$45,517	$43,604
Cost of sales	(21,386)	(17,803)	(17,137)
Gross profit	35,058	27,715	26,467
Distribution expenses	(5,876)	(4,543)	(4,259)
Sales and marketing expenses	(8,382)	(7,745)	(6,913)
Administrative expenses	(3,841)	(2,883)	(2,560)
Other operating income/(expenses)	854	732	1,032
Restructuring	(468)	(323)	(171)
Business and asset disposal	(39)	377	524
Acquisition costs business combinations	(155)	(448)	(55)
Impairment of assets	—	—	(82)
Judicial settlement	—	—	(80)
Profit from operations	17,152	12,882	13,904
Finance cost	(6,885)	(9,216)	(3,142)
Finance income	378	652	1,689
Net finance income/(cost)	(6,507)	(8,564)	(1,453)
Share of result of associates and joint ventures	430	16	10
Profit before tax	11,076	4,334	12,461
Income tax expense	(1,920)	(1,613)	(2,594)
Profit from continuing operations	9,155	2,721	9,867
Profit from discontinued operations	28	48	
Profit of the year	9,183	2,769	9,867
Profit from continuing operations attributable to:			
Equity holders of AB InBev	7,968	1,193	8,273
Non-controlling interest	1,187	1,528	1,594
Profit of the year attributable to:			
Equity holders of AB InBev	7,996	1,241	8,273
Non-controlling interest	$1,187	$1,528	$1,594

Anheuser-Busch InBev SA/NV Consolidated Income Statement (in Millions of US Dollars) - a multinational beverage company based in Belgium

	12 Months Ended		
	Dec. 31, 2017	Dec. 31, 2016	Dec. 31, 2015
Sales	$13,471.5	$6,597.4	$5,127.4
Excise taxes	(2,468.7)	(1,712.4)	(1,559.9)
Net sales	11,002.8	4,885.0	3,567.5
Cost of goods sold	(6,217.2)	(2,987.5)	(2,131.6)
Gross profit	4,785.6	1,897.5	1,435.9
Marketing, general and administrative expenses	(3,032.4)	(1,589.8)	(1,038.3)
Special items, net	(28.1)	2,522.4	(346.7)
Equity Income in MillerCoors	0	500.9	516.3
Operating income (loss)	1,725.1	3,331.0	567.2
Other income (expense), net			
Interest expense	(349.3)	(271.6)	(120.3)
Interest income	6.0	27.2	8.3
Other income (expense), net	(0.1)	(29.7)	0.9
Total other income (expense), net	(343.4)	(274.1)	(111.1)
Income (loss) from continuing operations before income taxes	1,381.7	3,056.9	456.1
Income tax benefit (expense)	53.2	(1,055.2)	(61.5)
Net income (loss) from continuing operations	1,434.9	2,001.7	394.6
Income (loss) from discontinued operations, net of tax	1.5	(2.8)	3.9
Net income (loss) including noncontrolling interests	1,436.4	1,998.9	398.5
Net (income) loss attributable to noncontrolling interests	(22.2)	(5.9)	(3.3)
Net income (loss) attributable to Molson Coors Brewing Company	$1,414.2	$1,993.0	$395.2

Molson Coors Brewing Company Consolidated Statement of Operations (in Millions of US Dollars) - US-based multinational brewing company

	Year Ended December 31	
	2016	2017
Sales	21,944	24,677
Cost of goods sold	(10,744)	(12,459)
Selling expense	(5,562)	(5,890)
General and administrative expense	(2,004)	(2,225)
Research and development expense	(333)	(342)
Other income (expense)	(278)	(219)
Recurring operating income	**3,022**	**3,543**
Other operating income (expense)	(99)	192
Operating income	**2,923**	**3,734**
Interest income on cash equivalents and short-term investments	130	151
Interest expense	(276)	(414)
Cost of net debt	(146)	(263)
Other financial income	67	137
Other financial expense	(214)	(312)
Income before tax	**2,630**	**3,296**
Income tax expense	(804)	(842)
Net income from fully consolidated companies	**1,826**	**2,454**
Share of profit of associates	1	109
Net income	1,827	2,563
Net income – Group share	**1,720**	**2,453**
Net income – Non-controlling interests	107	110

Groupe Danone Consolidated Income Statement (in Millions of Euros) - a French food manufacturer

Different orderings of chronological information are common!

Components and Format of the Income Statement (cont'd)

- Revenue (sales or turnover) generally refers to the amount charged for the delivery of goods or services in the ordinary activities of a business.
- Revenue is reported after adjustments (e.g., for cash or volume discounts or other reductions), and the term net revenue is sometimes used to indicate that revenue has been adjusted.
- In a comparative analysis, an analyst may need to reference information disclosed elsewhere in companies' annual reports to identify the appropriately comparable revenue amounts.

Components and Format of the Income Statement (cont'd)

- Expenses reflect outflows, depletions of assets, and incurrences of liabilities in the course of the activities of a business.
- Expenses may be grouped and reported in different formats, subject to some specific requirements.
- At the bottom of the income statement, companies report net income ("net earnings" or "profit or loss", "bottom line").
- Each company presents additional items below net income: information on how much of that net income is attributable to the company itself and how much is attributable to noncontrolling interests (minority interests).

Components and Format of the Income Statement (cont'd)

- The companies consolidate subsidiaries over which they have control.
- Consolidation means that they include all of the revenues and expenses of the subsidiaries, even if they own less than 100 percent.
- Noncontrolling interest represents the portion of income that "belongs" to the minority shareholders of the consolidated subsidiaries, as opposed to the parent company itself.
- Net income also includes gains and losses, which are increases and decreases in economic benefits, respectively, which may or may not arise in the ordinary activities of the business.

Components and Format of the Income Statement (cont'd)

- For example, when a manufacturing company sells its products, these transactions are reported as revenue, and the costs incurred to generate these revenues are expenses and are presented separately.
- However, if a manufacturing company sells surplus land that is not needed, the transaction is reported as a gain or a loss.
- The amount of the gain or loss is the difference between the carrying value of the land and the price at which the land is sold. (Example: See AB InBev "Business and asset disposal")

Components and Format of the Income Statement (cont'd)

- The definition of income encompasses both revenue and gains, and the definition of expenses encompasses both expenses that arise in the ordinary activities of the business and losses.
- Thus, net income (profit or loss) can be defined as:
  - a) income minus expenses,
  - b) revenue plus other income plus gains minus expenses,
  - c) revenue plus other income plus gains minus expenses in the ordinary activities of the business minus other expenses, and minus losses.
  - The last definition can be rearranged as follows: net income equals (i) revenue minus expenses in the ordinary activities of the business, plus (ii) other income minus other expenses, plus (iii) gains minus losses.

Components and Format of the Income Statement (cont'd)

- Under IFRS, certain items, such as revenue, finance costs, and tax expense, are required to be presented separately on the face of the income statement.
- Expenses may be grouped together either by their nature or function.
  - An example of a grouping by nature: grouping expenses such as depreciation on manufacturing equipment and depreciation on administrative facilities into a single line item called "depreciation".
  - An example of grouping by function: grouping together expenses into a category such as cost of goods sold, which may include labour and material costs, depreciation, some salaries (e.g., salespeople's), and other direct sales-related expenses

Components and Format of the Income Statement (cont'd)

- **Gross profit** is the amount of revenue available after subtracting the costs of delivering goods or services.
- When an income statement shows a gross profit subtotal, it uses a multi-step format rather than a single-step format.
- **Operating profit** results from deducting operating expenses, such as selling, general, administrative, and research and development expenses, from gross profit.
- Operating profit is sometimes referred to as EBIT (earnings before interest and taxes). However, operating profit and EBIT are not necessarily the same.

Revenue Recognition

- IFRS defines income as follows:
  - “Income is increases in economic benefits during the accounting period in the form of inflows or enhancements of assets or decreases of liabilities that result in increases in equity, other than those relating to contributions from equity participants.”
- In IFRS, the term “income” includes revenue and gains.
- Gains are similar to revenue, but they typically arise from secondary or peripheral activities rather than from a company's primary business activities (example: the sale of surplus equipment for more than its carrying value)
- Gains and losses may be considered part of operating activities (e.g., a loss due to a decline in the value of inventory) or may be considered part of non-operating activities (e.g., the sale of non-trading investments).

Revenue Recognition (cont'd)

- An important aspect concerning revenue recognition is that it can occur independently of cash movements.
  - For example, assume a company sells goods to a buyer on credit, so does not actually receive cash until some later time.
- A fundamental principle of accrual accounting is that revenue is recognized (reported on the income statement) when it is earned, so the company's financial records reflect revenue from the sale when the **risk and reward of ownership is transferred** (when the company delivers the goods or services).
  - Accounts receivable vs. Unearned revenue

Revenue Recognition (cont'd)

- Five steps in recognizing revenue:
    1. Identify the contract(s) with a customer
    2. Identify the separate or distinct performance obligations in the contract
    3. Determine the transaction price
    4. Allocate the transaction price to the performance obligations in the contract
    5. Recognize revenue when (or as) the entity satisfies a performance obligation

- Steps three and four address the amount, and step five addresses the timing of recognition.

Revenue Recognition (cont'd)

- Five Factors to consider when assessing whether the customer has obtained control of an asset at a point in time:
  - Entity has a present right to payment,
  - Customer has legal title,
  - Customer has physical possession,
  - Customer has the significant risks and rewards of ownership, and
  - Customer has accepted the asset.
- For a simple contract with only one deliverable at a single point in time, completing the five steps is straightforward. For more complex contracts, accounting choices can be less obvious.

Revenue Recognition (cont’d)

- Example 1:
  - Builder Co. enters into a contract with Customer Co. to construct a commercial building. Builder Co. identifies various goods and services to be provided, such as pre-construction engineering, construction of the building's individual components, plumbing, electrical wiring, and interior finishes. With respect to “**Identifying the Performance Obligation**,” should Builder Co. treat each specific item as a separate performance obligation to which revenue should be allocated?
  - The standard provides two criteria to determine if a good or service is distinct for purposes of identifying performance obligations.
    - First, the customer can benefit from the good or service either on its own or together with other readily available resources.
    - Second, the seller's “promise to transfer the good or service to the customer is separately identifiable from other promises in the contract.”
  - In this example, the second criterion is not met because it is the building for which the customer has contracted, not the separate goods and services. The seller will integrate all the goods and services into a combined output and each specific item should not be treated as a distinct good or service but accounted for together as a single performance obligation.

Revenue Recognition (cont’d)

- Example 2:
  - Builder Co.'s contract with Customer Co. to construct the commercial building specifies consideration of $1 million. Builder Co.'s expected total costs are $700,000. The Builder incurs $420,000 in costs in the first year. Assuming that costs incurred provide an appropriate measure of progress toward completing the contract, how much revenue should Builder Co. recognize for the first year?

  - The standard states that for performance obligations satisfied over time (e.g., under a long-term contract), revenue is recognized over time by measuring progress toward satisfying the obligation.
  - In this case, the Builder has incurred 60% of the total expected costs ($420,000/$700,000) and will thus recognize $600,000 (60% × $1 million) in revenue for the first year.

Revenue Recognition (cont’d)

- Example 3:
  - Assume that Builder Co.'s contract with Customer Co. to construct the commercial building specifies consideration of $1 million plus a bonus of $200,000 if the building is completed within 2 years. Builder Co. has only limited experience with similar types of contracts and knows that many factors outside its control (e.g., weather, regulatory requirements) could cause delay. Builder Co.'s expected total costs are $700,000. The Builder incurs $420,000 in costs in the first year. Assuming that costs incurred provide an appropriate measure of progress toward completing the contract, how much revenue should Builder Co. recognize for the first year?

  - A company is allowed to recognize variable consideration only if it can conclude that it will not have to reverse cumulative revenue in the future. In this case, Builder Co. does not recognize any bonuses in year one because it cannot reach a **non-reversible conclusion** given its limited experience with similar contracts and potential delays due to factors outside its control.

Revenue Recognition (cont'd)

- Example 4:
  - Assume all facts from Part 3. In the beginning of year two, Builder Co. and Customer Co. agree to change the building floor plan and modify the contract. As a result the consideration will increase by $150,000, and the allowable time for achieving the bonus is extended by 6 months. Builder expects its costs will increase by $120,000. Also, given the additional 6 months to earn the completion bonus, Builder concludes that it now meets the criteria for including the $200,000 bonus in revenue. How should Builder account for this change in the contract?
  - The company must update its transaction price and measure of progress.
  - Builder's total revenue on the transaction (transaction price) is now $1.35 million ($1 million original plus $150,000 new consideration plus $200,000 completion bonus).
  - Builder Co.'s progress toward completion is now 51.2% ($420,000 costs incurred divided by total expected costs of $820,000).
  - Based on the changes in the contract, the amount of additional revenue to be recognized is $91,200, calculated as (51.2% × $1.35 million) minus the $600,000 already recognized.
  - The additional $91,200 of revenue would be recognized as a "cumulative catch-up adjustment" on the date of the contract modification.

Revenue Recognition (cont’d)

- Example 5:
  - Assume a Company operates a website that enables customers to purchase goods from various suppliers. The customers pay the Company in advance, and orders are nonrefundable. The suppliers deliver the goods directly to the customer, and the Company receives a 10% commission. Should the Company report Total Revenues equal to 100% of the sales amount (gross) or Total Revenues equal to 10% of the sales amount (net)? Revenues are reported gross if the Company is acting as a Principal and net if the Company is acting as an Agent.

  - In this example, the Company is an Agent because it isn't primarily responsible for fulfilling the contract, doesn't take any inventory risk or credit risk, doesn't have discretion in setting the price, and receives compensation in the form of a commission.
  - Because the Company is acting as an Agent, it should report only the commission amount as its revenue.

International Financial Statement Analysis

Chapter 3: Understanding Income Statements – Part 2

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Expense Recognition

- The definition of expenses encompasses losses as well as expenses that arise in the ordinary course of the enterprise's activities:
  - cost of sales, wages and depreciation, etc.
- They usually take the form of outflows or depletion of assets such as cash and cash equivalents, inventory, property, plant, and equipment.
- Losses represent decreases in economic benefits and, as such, they are no different in nature from other expenses:
  - resulting from disasters such as fire and flood, arising from the disposal of non-current assets, etc.

Expense Recognition (cont'd)

- Similar to the issues with revenue recognition, expense recognition would not be an issue in a simple hypothetical scenario.
- In practice, however, as with revenue recognition, determining when expenses should be recognized can be somewhat more complex.
- **Matching Principle**
  - In general, a company recognizes expenses in the period that it consumes (i.e., uses up) the economic benefits associated with the expenditure, or loses some previously recognized economic benefit.

Expense Recognition (cont'd)

- Under matching, a company recognizes expenses (e.g., cost of goods sold) when associated revenues are recognized, thereby matching expenses to revenues.
- However, in practice, it is more likely that some of the current period's sales are made from inventory purchased in a previous period or previous periods.
- Matching requires that a company recognize the cost of goods sold in the same period as the revenue from the sale of those goods.
- **Period costs**, expenditures that are less directly matched to revenues, are recognized in the period in which a company incurs the expenditure or liability to pay. (e.g. Administrative expenses)

Expense Recognition - Example (specific identification method)

- Kahn Distribution Limited (KDL), a hypothetical company, purchases inventory items for resale. At the beginning of 2018, Kahn had no inventory on hand. During 2018, KDL had the following transactions:

Inventory Purchases		
First quarter	2,000	units at $40 per unit
Second quarter	1,500	units at $41 per unit
Third quarter	2,200	units at $43 per unit
Fourth quarter	1,900	units at $45 per unit
Total	7,600	units at a total cost of $321,600

- KDL sold 5,600 units of inventory during the year at $50 per unit and received cash. KDL determines that there were 2,000 remaining units of inventory and specifically identifies that 1,900 were those purchased in the fourth quarter and 100 were purchased in the third quarter. What are the revenue and expenses associated with these transactions during 2018? **(specific identification method)**

Expense Recognition - Example (solution)

- The revenue for 2018 would be $280,000 (5,600 units × $50 per unit). Initially, the total cost of the goods purchased would be recorded as inventory (an asset) for $321,600. During 2018, the cost of the 5,600 units sold would be expensed (matched against the revenue) while the cost of the 2,000 remaining unsold units would remain in inventory as follows:

Cost of Goods Sold		
From the first quarter	2,000 units at $40 per unit =	$80,000
From the second quarter	1,500 units at $41 per unit =	$61,500
From the third quarter	2,100 units at $43 per unit =	$90,300
Total cost of goods sold		$231,800
Cost of Goods Remaining in Inventory		
From the third quarter	100 units at $43 per unit =	$4,300
From the fourth quarter	1,900 units at $45 per unit =	$85,500
Total remaining (or ending) inventory cost		$89,800

Revenue	$280,000
Cost of goods sold	231,800
Gross profit	$48,200

Expense Recognition (cont'd)

- It is generally not feasible to specifically identify which items were sold and which remain on hand, so accounting standards permit the assignment of inventory costs to costs of goods sold and to ending inventory using cost formulas.
  - **FIFO method:** the oldest goods purchased (or manufactured) are assumed to be sold first, and the newest goods purchased (or manufactured) are assumed to remain in inventory.
  - **The weighted-average cost method** assigns the average cost of goods available for sale to the units sold and to those remaining in inventory.
  - **LIFO method:** the newest goods purchased (or manufactured) are assumed to be sold first, and the oldest goods purchased (or manufactured) are assumed to remain in inventory. (Not permitted under IFRS)

Expense Recognition - Example (FIFO, Average method, LIFO)

- FIFO:
  - It turns out that the items specifically identified as sold in Example 1 were also the first items purchased, so under FIFO, the cost of goods sold would be $231,800.
- Average method:
  - For KDL, the weighted average cost per unit would be: $321,600/7,600 units = $42.3158 per unit
  - Cost of goods sold using the weighted average cost method would be: 5,600 units at $42.3158 = $236,968
  - Ending inventory using the weighted average cost method would be: 2,000 units at $42.3158 = $84,632
- LIFO:
  - 1,900 units at $45 per unit + 2,200 units at $43 per unit + 1,500 units at $41 per unit = $241,600

Expense Recognition (cont'd)

Method	Description	Cost of Goods Sold When Prices Are Rising, Relative to Other Two Methods	Cost of Goods Sold When Prices Are Rising, Relative to Other Two Methods
FIFO	Costs of the earliest items purchased flow to cost of goods sold first	Lowest	Highest
LIFO	Costs of the most recent items purchased flow to cost of goods sold first	Highest	Lowest
Weighted Average Cost	Averages total costs over total units available	Middle	Middle

Expense Recognition (cont'd)

- Issues in Expense Recognition
  - Doubtful Accounts
    - Some customers will ultimately default on their obligations (i.e., fail to pay). Under the matching principle, a company is required to record an estimate of how much revenue will ultimately be uncollectible.
  - Warranties
    - Under the matching principle, a company is required to estimate the amount of future warranty expenses, recognize an estimated warranty expense in the period of sale, and update the expense as indicated by experience over the life of the warranty.
  - Depreciation and Amortization
    - The costs of most long-lived assets are allocated over the period of time during which they provide economic benefits.
    - The two main types of long-lived assets whose costs are not allocated over time are land and intangible assets with indefinite useful lives.

Expense Recognition (cont'd)

- **Depreciation** is the process of systematically allocating costs of long-lived assets over the period during which the assets are expected to provide economic benefits.
- **"Depreciation"** is the term commonly applied to this process for physical long-lived assets (such as plant and equipment), and **"amortization"** is the term commonly applied to this process for intangible long-lived assets with a finite useful life (such as mailing list, patent, copyright).
- IFRS allow two alternative models for valuing property, plant, and equipment:
  - **The cost model:** the asset is reported at its cost less any accumulated depreciation. (widely used)
  - **The revaluation model:** the asset is reported at its fair value. (not permitted under U.S. GAAP)
- Depreciation methods: the straight-line method, diminishing balance method (accelerated depreciation), and the units of production method.

Expense Recognition (cont'd)

- **The straight-line method** allocates the cost of long-lived assets, less the estimated residual value, evenly over the estimated useful life of the asset.
- Calculating depreciation and amortization requires two significant estimates:
  - The estimated useful life of an asset, and
  - The estimated residual value (also known as "salvage value") of an asset. Under IFRS, the residual value is the amount that the company expects to receive upon sale of the asset at the end of its useful life.
- Using the straight-line method of depreciation, annual depreciation expense is calculated as:

$$\frac{Cost - Residual\ Value}{Estimated\ Useful\ Life}$$

Expense Recognition – Example (straight-line method)

- Assume the cost of an asset is $10,000.
  - If the residual value of the asset is estimated at $0 and its useful life is estimated at 5 years, the annual depreciation expense under the straight-line method would be ($10,000 − $0)/5 years = $2,000.
  - In contrast, holding the estimated useful life of the asset constant at 5 years while increasing its estimated residual value to $4,000 would result in an annual depreciation expense of only $1,200 [calculated as ($10,000 − $4,000)/5 years].
  - Alternatively, holding the estimated residual value at $0 but increasing the estimated useful life of the asset to 10 years would result in an annual depreciation expense of only $1,000 [calculated as ($10,000 − $0)/10 years].

Estimated Useful Life (Years)	Estimated Residual Value					
	0	1,000	2,000	3,000	4,000	5,000
2	5,000	4,500	4,000	3,500	3,000	2,500
4	2,500	2,250	2,000	1,750	1,500	1,250
5	2,000	1,800	1,600	1,400	1,200	1,000
8	1,250	1,125	1,000	875	750	625
10	1,000	900	800	700	600	500

Expense Recognition – Example (accelerated methods)

- Assume the cost of computer equipment was $11,000, the estimated residual value is $1,000, and the estimated useful life is five years.
- The first step is to determine the straight-line rate, the rate at which the asset would be depreciated under the straight-line method. This rate is measured as 100 percent divided by the useful life, or 20 percent for a five-year useful life.
- The next step is to determine an acceleration factor that approximates the asset's wear pattern. Common acceleration factors are 150 percent and 200 percent (double declining balance depreciation). This rate is then applied to the remaining undepreciated balance of the asset each period (known as the net book value).
- At the beginning of the first year, the net book value is $11,000. Depreciation expense for the first full year of the asset's use would be 40 percent of $11,000, or $4,400.
- At the beginning of Year 2, the net book value is measured as

Asset cost	$11,000
Less: Accumulated depreciation	(4,400)
Net book value	$6,600

Expense Recognition – Example (accelerated methods)(cont'd)

- For the second full year, depreciation expense would be $6,600 × 40 percent, or $2,640. At the end of the second year (i.e., beginning of the third year), a total of $7,040 ($4,400 + $2,640) of depreciation would have been recorded. So, the remaining net book value at the beginning of the third year would be

Asset cost	$11,000
Less: Accumulated depreciation	(7,040)
Net book value	$3,960

- For the third full year, depreciation would be $3,960 × 40 percent, or $1,584. At the end of the third year, a total of $8,624 ($4,400 + $2,640 + $1,584) of depreciation would have been recorded. So, the remaining net book value at the beginning of the fourth year would be

Asset cost	$11,000
Less: Accumulated depreciation	(8,624)
Net book value	$2,376

- For the fourth full year, depreciation would be $2,376 × 40 percent, or $950. At the end of the fourth year, a total of $9,574 ($4,400 + $2,640 + $1,584 + $950) of depreciation would have been recorded. So, the remaining net book value at the beginning of the fifth year would be

Asset cost	$11,000
Less: Accumulated depreciation	(9,574)
Net book value	$1,426

- For the fifth year, if deprecation were determined as in previous years, it would amount to $570 ($1,426 × 40 percent). However, this would result in a remaining net book value of the asset below its estimated residual value of $1,000. So, instead, only $426 would be depreciated, leaving a $1,000 net book value at the end of the fifth year.

Asset cost	$11,000
Less: Accumulated depreciation	(10,000)
Net book value	$1,000

Expense Recognition (cont’d)

- Implications for Financial Analysis
  - A company's estimates for doubtful accounts, warranty expenses, choice of depreciation or amortization method, estimates of assets' useful lives, and estimates of assets' residual values can affect reported net income.
  - If, for example, a company shows a significant year-to-year change in its estimates of uncollectible accounts as a percentage of sales, warranty expenses as a percentage of sales, or estimated useful lives of assets, the analyst should seek to understand the underlying reasons.
  - If two companies in the same industry have dramatically different estimates for uncollectible accounts as a percentage of their sales, warranty expenses as a percentage of sales, or estimated useful lives as a percentage of assets, it is important to understand the underlying reasons.
  - Information about a company's accounting policies and significant estimates is described in the notes to the financial statements and in the management discussion and analysis section of a company's annual report.